Ben Bubnick

Master Career History

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Career Overview

Comprehensive working record spanning healthcare data consulting, data engineering, analytics, software development, scientific computing, higher education, academic research, and community service. Combines hands-on delivery with technical leadership, documentation, teaching, stakeholder engagement, and quality-focused practices. Maintained as a source for targeted resumes, biographies, applications, and future career pivots.

Skills and Domains

Data Engineering and Software: SQL, Python, Ruby, Snowflake, dbt, SQLMesh, Spark, Hadoop/HDFS, Hive, Impala, Kafka, PostgreSQL, Oracle, C#, VB.NET, C++, FORTRAN, ETL, ELT, data warehouses, data lakes
Cloud and Delivery: AWS, Azure, Docker, Git/GitHub, GitHub Actions, Jenkins, Azure DevOps, Jira, Confluence, CI/CD, technical documentation, mentoring
Healthcare and Analytics: Clinical, payer, eligibility, revenue-cycle, and financial data; HL7, FHIR, EDI/837, HEDIS, EHR/EMR, HIPAA, HITRUST, pandas, Jupyter, Power BI, Tableau, statistical modeling
Science and Education: Spectroscopy, photometry, optical measurement, radiative-transfer modeling, Fourier analysis, Monte Carlo methods, numerical integration and interpolation, physics and astronomy instruction, curriculum development, research publication
Earlier Scientific and Creative Tools: R, PERL, IDL, IRAF, Mathematica, MATLAB, LabVIEW, Maple, PDL, JMP, SAS/GRAPH, AutoCAD, Adobe Illustrator, Adobe Photoshop, GIMP, LaTeX

Professional Experience

HHAeXchange Oct. 2025 - Jun. 2026
Healthcare Data Integration Consultant - Contract via Veersa Technologies
Technologies: Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase, GitHub Actions, FHIR/Aidbox
Led healthcare source analysis, data-quality investigation, and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use.

- Engineered unified and analytics Snowflake/dbt models spanning 55 tables across two organizations and two source-system configurations, reducing client onboarding time by 67%.
- Implemented reusable SQL/Jinja transformation logic for lifecycle states, null handling, sentinel dates, and source-system variations across organization, office, caregiver, and patient domains.
- Created or materially redesigned 30 automated SQL/dbt data-quality tests covering nulls, duplicates, referential integrity, business rules, lifecycle logic, and source-to-target reconciliation.
- Translated ambiguous business requirements into documented mappings, acceptance criteria, onboarding standards, and 12 data-quality investigation runbooks.

Adonis Mar. 2025 - Jul. 2025
Healthcare Revenue Cycle Data Consultant - Contract via Health Data Movers
Technologies: AWS S3, Snowflake, SQLMesh, SQL, Python, Docker
Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding, analytics, and reporting.

- Built, documented, validated, and maintained 54 SQLMesh models supporting revenue-cycle analytics for a provider serving 4,598 families across 113 locations and processing 86,388 annual authorizations.
- Implemented automated row-count, uniqueness, and not-null audits across 54 models, achieving a 100% pass rate across 117+ configured audit applications at final delivery.
- Diagnosed undocumented source-key versioning and implemented composite business grains with SHA-256 surrogate keys across 10 charge and payment models, eliminating duplicate-key failures.
- Authored implementation playbooks, technical handoff documentation, and a QA runbook for a repeatable healthcare-data validation workflow.

GrowData Analytics Sep. 2024 - Oct. 2025
Independent Data Automation and Analytics Consultant
Technologies: Ruby, RSpec, OCR, LLM APIs, audit logging, automated testing

- Built and validated a preproduction document-to-data MVP that converted semi-structured invoices into auditable transaction records, achieving 95% field-level accuracy across a stratified 500-invoice test suite.
- Instrumented the MVP with automated tests, audit logging, retry controls, and confidence routing, achieving 88% straight-through processing and under 1% system failure.

Arkos Health Jun. 2023 - Jan. 2025
Senior Data Analyst
Technologies: AWS S3, Athena, Redshift, Glue, dbt, PySpark, Docker, Jupyter, Power BI, Tableau

- Architected a greenfield AWS data lake integrating 18 source systems through approximately 200 scheduled data assets and 200 dbt models, supporting six analysts across six state markets.
- Integrated Jupyter-based analytics and authored platform, architecture, and operating documentation, enabling six analysts to independently use the governed data lake for recurring, executive, and BI reporting.
- Instrumented scheduled pipelines with monitoring, alerting, and more than 10 failure classifications; authored three AWS/dbt incident runbooks supporting rapid detection and recovery.

Merative Jul. 2022 - Mar. 2023
Senior Data Scientist
Technologies: Azure Data Factory, Blob Storage, Functions, SQL Database, Synapse Analytics, Python, SQL, Azure DevOps

- Engineered a seven-layer Python and SQL migration-acceptance suite covering partition reconciliation, schema compatibility, null handling, duplicates, field-level comparisons, aggregates, and business rules, achieving exact HDFS-to-Azure fidelity.
- Led HIPAA-compliant deidentification validation across demographic, encounter, and coded clinical data representing approximately 300 million patient lives, achieving a 100% pass rate before sign-off.

IBM Jun. 2015 - Jul. 2022
Advanced through five roles delivering healthcare analytics, data architecture, client integration, data-quality improvement, and large-scale migration validation.
Technologies: Spark, Hadoop/HDFS, Hive, Impala, Kafka, Python, SQL, PostgreSQL, Docker, Jenkins, Tableau, HL7

Senior Data Scientist Nov. 2021 - Jul. 2022

- Developed a surgical-scheduling optimizer and CNN/RNN case-card prediction proof of concept, automating case-card generation and resource allocation across operating rooms, time slots, and personnel.
- Engineered a Spark, Hive, and Impala analytics framework powering 50+ prior-authorization lag-time and approval-bottleneck reports with daily operational and periodic regulatory and financial snapshots.

Senior Data Architect Dec. 2020 - Nov. 2021

- Architected Spark and Hadoop pipelines integrating more than 100 billion structured and semi-structured clinical records, supporting four EPM applications, 30+ risk models, and hundreds of measures and registries.
- Replaced static mappings with metadata-driven instruction sets across approximately 200 source integrations, standardizing EDI batch schemas while enabling Python/Jupyter lineage tracing and end-to-end validation gates.

Data Architect Nov. 2017 - Dec. 2020

- Designed SQL validation frameworks and client-review gates across approximately 100 integrations, reducing Jira-tracked validation rework by 25% and mapping rework by 45%.
- Led a multi-organization healthcare integration spanning 70+ EHR sources, implementing provider-scoped patient filtering to enforce data-sharing rules and coordinating delivery with engineering teams.

Software Developer, Data Platform Sep. 2016 - Nov. 2017

- Engineered an HL7 v2/v3 parsing pipeline in Pig for approximately 30 internal users, replacing a Mirth-based workflow and eliminating $30,000 in annual vendor costs.
- Built a Python, Ruby, and SQL extraction tool that reduced total client-integration effort by 28%, measured through eliminated story points and lower average engineering hours per implementation.

Data Scientist Nov. 2015 - Sep. 2016

- Directed technical delivery across 36 client data integrations, coordinating 12 remote contractors and establishing code-review, QA, and issue-resolution standards.
- Authored 125+ technical knowledge-base articles, approximately 30% of the team's published content, to standardize integration development and troubleshooting practices.

Explorys Jun. 2015 - Nov. 2015
Data Science Consultant - Contract via Randstad USA
Technologies: Hadoop/HDFS, Ruby, Pig, React, Node.js, JavaScript, SQL, Oracle, Jenkins
Supported clinical-data integration and population analytics across more than 10 client data warehouses.

- Diagnosed Hadoop and Oracle production-ingestion failures and implemented Jenkins-backed scheduling and recovery controls across 10+ client data warehouses.

AmTrust Financial Services Mar. 2014 - Jun. 2015
Application Development Consultant - Contract via TEKsystems
Technologies: VB.NET, ASP.NET, C#, Web APIs, CSS, JavaScript, HTML5, SQL, PL/SQL
Developed ISO-based web forms, multi-tier MVC components, Web APIs, and user interfaces for commercial insurance policy software.

- Led development work for insurance policy software, a new product offering, and the business integration of acquisitions.
- Analyzed more than 65 rating forms to identify and resolve performance bottlenecks.
- Introduced C# unit testing for multi-tier MVC components and helped train new staff while coordinating daily work allocation.

PPG Industries Apr. 2012 - Aug. 2013
Color Scientist
Technologies: VB.NET, C#, SQL, C++, FORTRAN, JMP, spectrophotometry, numerical methods
Developed software and radiative-transfer models for automotive paint formulation, spectroscopic profile analysis, and technology-risk assessment.

- Improved a color-matching algorithm by 40% using numerical integration methods.
- Built cross-instrument correlation software in VB.NET and JMP to improve measurement consistency and support statistical risk analysis.
- Debugged a VB.NET and FORTRAN optimization routine for a numerical radiative-transfer solution.
- Wrote a Visual C++ and VB.NET DLL wrapper for a loaned spectroscopic instrument and developed formulation methods for an automated color-matching facility.

Teaching and Academic Research

Lorain County Community College Aug. 2016 - May 2019
Adjunct Professor - Taught physics laboratory and astronomy lecture courses.
Miami University Aug. 2010 - Jun. 2011
Adjunct Professor - Developed and taught Physics and Society, introductory physics, and Astronomy and Space Physics courses for more than 100 students.
Cincinnati State Community College Feb. 2011 - Jun. 2011
Adjunct Professor - Taught technical physics for remedial students; wrote laboratory instructions and coordinated measurement, analysis, and technical-report work for five student teams.
University of Cincinnati Oct. 2008 - Jun. 2010
Graduate Teaching Assistant - Administered approximately 60 algebra- and calculus-based physics laboratories for more than 100 engineering and non-physics students; taught measurement, Microsoft Excel, and DataStudio.
University of Cincinnati Oct. 2009 - Dec. 2009
Engineering Teaching Assistant - Prepared and delivered Mathematica lectures for advanced aerospace-engineering students.
University of Cincinnati Jun. 2006 - Dec. 2006
Supplemental Instruction Leader - Led peer-learning sessions of up to 12 students in calculus- and algebra-based introductory physics.
University of Cincinnati Jan. 2006 - May 2006
Undergraduate Teaching Assistant - Supported group recitations in freshman engineering physics through demonstrations, short lectures, and individual assistance.
University of Cincinnati Learning Communities Program Jun. 2006 - Aug. 2008
Peer Leader - Connected first-year students with faculty, academic advisers, program staff, and campus resources; mentored their transition to university life and introduced academic tools, time-management, study, and community-building skills.
Private Tutoring Jun. 2006 - Aug. 2008
Private Tutor - Tutored honors and introductory physics students in course concepts and problem-solving techniques.
Across these institutions, used interactive, peer-led, and just-in-time teaching methods; produced more than 700 hours of recorded lectures and laboratory material; and exceeded evaluation benchmarks by 16–17% year over year.
University of Cincinnati Sep. 2006 - Sep. 2010
Research Assistant, Department of Physics - Conducted near-infrared astronomical research using observations from the 0.8–5.4 micrometer SpeX spectrograph at the NASA Infrared Telescope Facility.

- Used photometric data to estimate the distance, extinction, and physical properties of open star clusters and help map the structure of the Milky Way.
- Located more than 100 stars in each observed field and calculated radial velocities using physical and Fourier cross-correlation methods.
- Performed spectroscopy, photometry, classification, data reduction, signal analysis, and noise reduction using IRAF, IDL, Mathematica, PERL, and related scientific tools.
- Volunteered 120 hours of quality control for the Cambridge Astronomical Survey Unit Variable Stars database.
- Operated the SpeX spectrograph as a visiting astronomer at the NASA Infrared Telescope Facility on Mauna Kea in

November 2007.
- Published research on near-infrared astrophysics and received the Mary J. Hanna Research Fellowship in 2010.

Ohio University Oct. 2003 - May 2004
Undergraduate Student Researcher, Printmaking - Used a competitive undergraduate research grant to develop a lower-toxicity photo-etch process that replaced conventional photochemical materials with toner and a wintergreen-based solvent.

Earlier Employment

University of Cincinnati Geology Math Physics Library Mar. 2006 - Sep. 2008
Student Librarian - Assisted patrons with research and reference needs, maintained Library of Congress-classified materials, conducted inventory, and located missing items.
FranklinCovey Jun. 2005 - Apr. 2006
Productivity Consultant and Sales Associate - Advised customers on planning products and workshops, used detailed product knowledge to meet client needs, and supported cash and credit transactions.
Retail Food Service and Manual Work 1995 - 2010
Various roles - Worked as a Target cashier; Dillard's sales associate; Aladdin's Eatery dishwasher; Cravings Cafe and Take 5 Coffee Company barista; Pickle Bill's Lobster House cook; and Pier W dishwasher and cook. Other early work included retail sales, coffee-shop management, telemarketing, industrial labor, FedEx loading, interior painting, cement work, yard work, swim instruction, and grocery and restaurant service.
The customer-facing and operations roles included opening and closing, training coworkers, ordering and receiving inventory, stocking and heavy lifting, handling cash and credit transactions, preparing food and espresso drinks, and resolving customer concerns during high-volume service.

Volunteer and Community Experience

HSEA Feb. 2025 - May 2025
ETL Brigade Volunteer - Built a reusable PostgreSQL model for community-health analytics covering claims, visits, and eligibility.
CSforAll Oct. 2018 - May 2019
Instructor - Supported computer-science instruction for high-school students, with emphasis on learners without reliable home computer or Internet access.
HIT in the CLE Mar. 2017 - Mar. 2019
Coach - Used a data-science competition to introduce students to health-information-technology careers and support the regional talent pipeline.
Code for America Jan. 2015 - Jun. 2015
ETL Guild Volunteer - Applied data-science tools to improve and expand public-record access.
Heights Observer, Cleveland Heights, Ohio Feb. 2012 - May 2014
Volunteer Editor - Edited community-newspaper content in support of local civic engagement and an independent free press.
The News Record, University of Cincinnati Feb. 2008 - Sep. 2010
Editorial Contributor - Wrote and edited content for the university's student newspaper.
TCP World Academy Sep. 2008 - May 2009
Tutor - Helped students connect classroom learning with participatory society-simulation activities.
Project Astro at Ethel M Taylor Academy 2008
Volunteer Astronomer - Supported a national program connecting astronomers with educators to strengthen astronomy and physical-science instruction.
Reach Out on Campus at Ohio University Sep. 2000 - Jun. 2004
Volunteer - Participated in campus-based community service.

Leadership and Professional Affiliations

Detroit Color Council 2012 - 2013
Member
American Institute of Aeronautics and Astronautics Cincinnati Student Chapter 2009 - 2011
Vice President in 2010; participated in Students for the Exploration and Development of Space in 2009.
University of Cincinnati Astronomy Club 2008 - 2010
Founding member in 2008, treasurer from 2008 to 2009, and secretary from 2009 to 2010.
Society of Physics Students Cincinnati Chapter 2005 - 2008
Secretary from 2005 to 2006 and treasurer from 2006 to 2007.
Sigma Theta Epsilon Beta Gamma Chapter 2005 - 2008
Held secretary, treasurer, chaplain, and publicity responsibilities.
Additional Academic Activities 2006 - 2009
Participated in the Large Synoptic Survey Telescope seminar group, UC SPARKS, and the University of Cincinnati Mountaineering Club.
Athens Wargamers Jan. 2001 - Jun. 2004
Member; served as treasurer during the 2002–2003 academic year.

Selected Independent Projects

farm-invoice-parser 2025
Ruby, RubyLLM, OCR, and model APIs; converts handwritten farm invoices into structured transaction records.
healthcare-dbt-demo 2025
Runnable Data Vault 2.0 demonstration for healthcare claims data using dbt and DuckDB/PostgreSQL.
tuva-postgres 2025
Python and PostgreSQL utilities for converting and validating Tuva seed data.
HL7-ingestion 2023
Java and Docker healthcare-interoperability project converting HL7 messages into longitudinal patient records.
quality_measures_engine 2023
Python project for healthcare quality-measure processing and evaluation workflows.
DETL-design-challenge 2023
Health-data transformation and loading design demonstrating ETL automation and validation.

Publications and Research Outputs

Reproducible Postgres Load of Tuva Seed Datasets Aug. 2025
Reusable load and validation workflow for Tuva healthcare seed datasets.
Surgical Scheduling Optimization and Procedure Duration Prediction Dec. 2020
Optimization and prediction work improving on fixed-ratio methods for estimating total procedure time.
Process and Requirements for CCD Data Integration Oct. 2017
Processes and requirements for addressing challenges in CCD data curation.
AI Paint Formula Prediction from Spectroscopic Measurements May 2015
Naive Bayesian approach to color and effects matching for complex automotive paint formulas.
Characterization of Surface Coatings Using RT Models for Color Matching Feb. 2013
Approximation of a Chandrasekhar radiative-transfer model applied to atmospheric scattering and color matching.
Massive Stellar Clusters in the Disk of the Milky Way Galaxy Dec. 2010
Master's thesis presenting spectroscopic and photometric analysis of two open clusters in the Galactic plane.
The B Stars of Westerlund 1 2009 - 2010
Near-infrared stellar-classification research manuscript prepared with M. M. Hanson; listed in contemporaneous resumes as in preparation.
VizieR Online Data Catalog: JHK Photometry in Cluster [BDS2003] 107 May 2008
Near-infrared photometric catalog associated with observations of the Galactic cluster.
Near-Infrared Spectroscopy of Candidate Members of Galactic Cluster [BDS2003] 107 Feb. 2008
Classification spectra for nine candidate cluster members, published in Publications of the Astronomical Society of the Pacific, volume 120, page 864.
A Novel Method for Low-Toxic Photo-Etch Printing Fall/Winter 2003
Published by the Mid America Print Council (MAPC), Volume 11, Number 2; presented undergraduate research applying aquatint etching techniques to photo-etch procedures.

Speaking and Knowledge Sharing

AI Hackathon: Surgery Scheduling Optimization Dec. 2020
Presented an approach using historical utilization and current wait-list information to predict surgery duration.
REACH Initiative: IBM Watson Care Insights 2019
Presented the use of patient records and real-time HL7 feeds to integrate insights into clinical workflows.
Content and Data Analytics Learning Exchange 2018
Introduced machine-learning implementation using real-world health-data analytics.
Innovation Research Interchange Fall Networks Conference Sep. 2018
Presented natural-language processing for extracting insight from unstructured data lakes.
Cross-Team Training Series 2016
Led a seven-part series on Hadoop ingestion and data-lake concepts.

Professional Recognition

Special Recognition, Health Data Movers - client delivery 2025
Arkos Health BI Star - Salesforce Trailhead Expeditioner 2023
Manager Recognition for Transition Leadership, Merative - systems migration 2022
Gold Champion, IBM - 1,477 classroom hours and 43 certificates 2018 - 2022
Lab Services Award, IBM - cross-team delivery coordination 2018
Eminence and Excellence Award, IBM - data-curation tool development 2017
Color Lab Automation Award, PPG Industries - expanded reporting 2012
Mary J. Hanna Research Fellowship, University of Cincinnati 2010
Academic Scholarships and Honors 2000 - 2011
University Graduate Scholarships for physics study from 2008 to 2010 and aerospace-engineering study in 2011; University of Cincinnati Dean's List from 2005 to 2007; Ohio University Provost's Undergraduate Research Grant in 2003 and Dean's List from 2002 to 2004; Ohio Proficiency Scholarship in 2000.

Education and Credentials

IBM Data Science Professional Certificate, IBM, 2022
Certificate of Specialization, Data Science, Johns Hopkins University, 2015
Graduate study, Aerospace Engineering, University of Cincinnati, 2011 - Computational Fluid Dynamics; degree not completed
M.S. Physics, Astrophysics concentration, University of Cincinnati, Dec. 2010 - thesis option; GPA 3.3/4.0
B.F.A. Printmaking, Ohio University, Aug. 2004 - major GPA 3.6/4.0
Additional coursework: Windows Forms; quantitative finance; information risk; computer networks; computational data analysis; AI planning; astrobiology; critical thinking